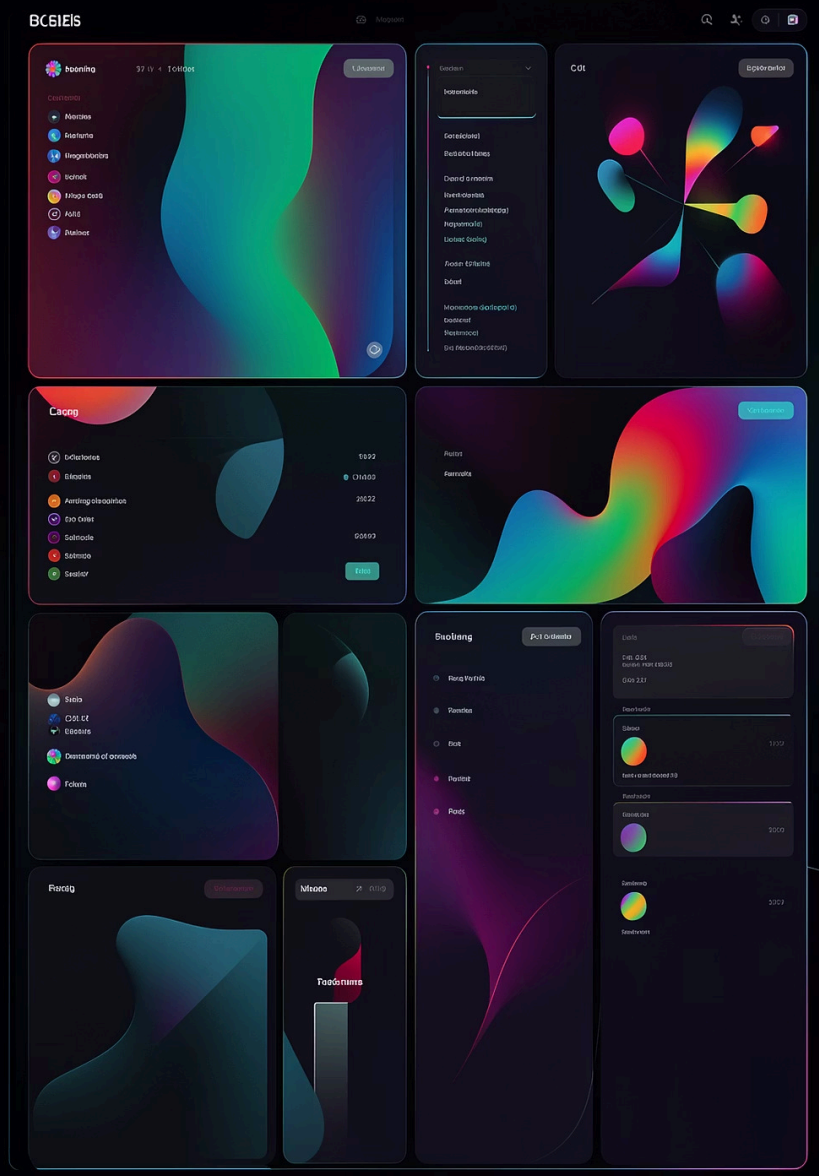
NexusOS — Insurance Agent OS

Engineering Context Document | Confidential | April 2026

> "The last software before AI fully automates insurance sales."



What We're Building

NexusOS is a **vertical-integrated insurance agent operating system** — not a generic CRM. Built specifically for life insurance brokerages with 2,500 independent agents.

CRM Pipeline
GHL-inspired stages, drag-drop, auto-triggered SMS/email workflows

Power Dialer + AI Coaching
Real-time AI coaching during live calls — **THE MOAT**

Live Transcription
Medical underwriting extraction from live call audio

AI Carrier Ranking
Auto-quoting across carrier APIs based on extracted health profile

Commission Dashboard
Chargeback tracking, renewals, lapse alerts

Agent Leaderboard
KPIs, performance metrics, campaign management

Calendar Integration
Google/Outlook sync with TCPA-compliant infrastructure

Market & Business

2,500

Independent Agents

First client: a life insurance brokerage at full adoption

$2.5M

MRR Potential

At ~$1,000/agent/month at full adoption

Phase 2 replicates the same skeleton for **healthcare insurance** with different carriers. They currently use LeadChampPro (a GHL skin) — we have login access to study their workflows.

Exit Thesis

Build the dominant vertical platform now, sell before full automation makes the human agent obsolete.

⚠ Window of opportunity is finite. The race is on.

Pricing Model

~$1,000/agent/month — enterprise SaaS pricing justified by the depth of vertical integration and AI moat.

The Moat — What Generic Tools Cannot Do

Generic AI coaching tools (Cluely, Gong, Chorus) overlay Zoom/Meet with general suggestions. NexusOS goes far deeper.

Insurance-Specific Medical Triggers

Client says *"I have diabetes"* → AI immediately prompts: **"Type 1 or Type 2? A1C level? When diagnosed? Insulin or metformin?"** — not generic follow-ups.

Live Underwriting Fact Panel

Structured underwriting facts update in real-time as the conversation progresses. Carriers re-rank **LIVE** as new health facts emerge during the call.

Post-Call Quote Comparison

Generates a quote comparison with top 3 carriers, premiums, and explicit **REASONS** — all deterministic, auditable, and E&O-defensible.

No RAG. No Hallucination.

ALL ranking logic uses deterministic TypeScript rules — not retrieval-augmented generation. Every decision is auditable and legally defensible.

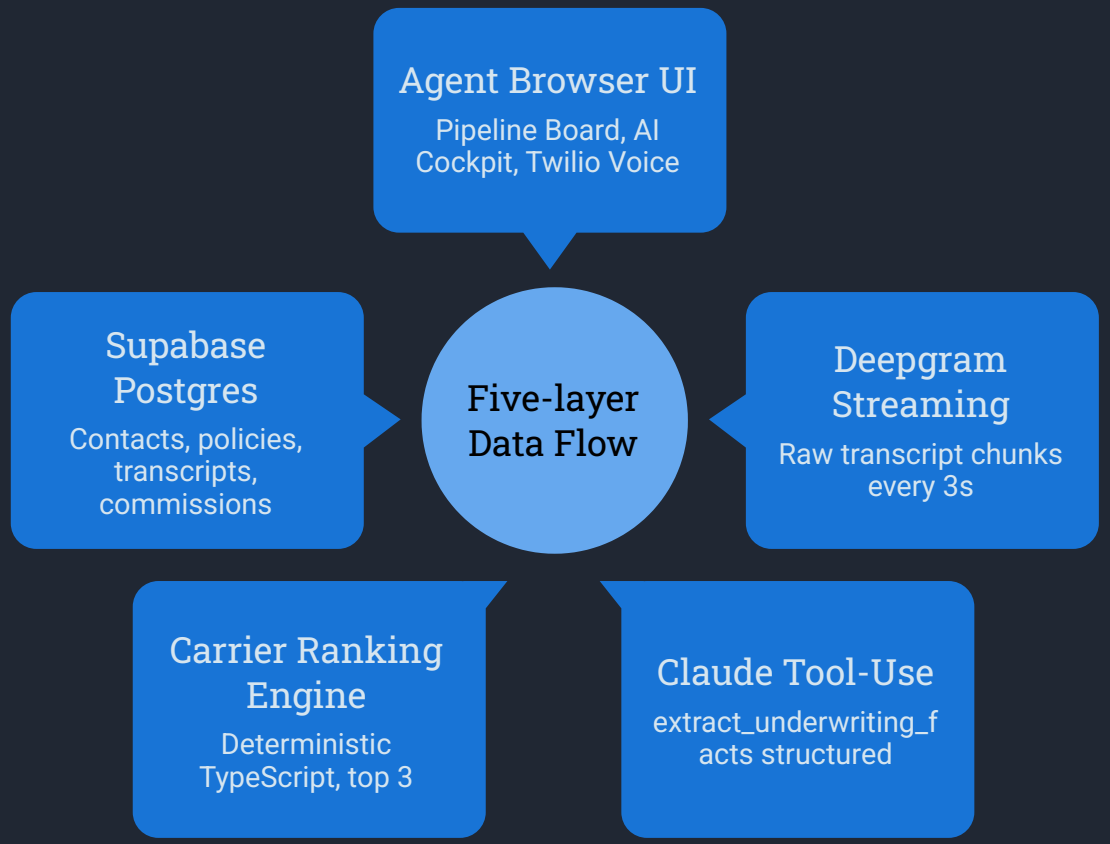
Tech Stack

Layer	Technology
Framework	Next.js 15 (App Router, React 19, TypeScript strict)
UI	shadcn/ui + Tailwind CSS v4 + Framer Motion
Database	Supabase (Postgres + RLS + Realtime + Edge Functions + pgvector)
Auth	Supabase Auth → SSO (Microsoft Entra / Google Workspace)
Voice	Twilio Voice SDK (browser-based softphone)

Layer	Technology
Transcription	Deepgram Streaming API
AI Extraction	Claude API (Sonnet, streaming tool-use)
Workflows	n8n (self-hosted)
Hosting	Vercel + VPS (Traefik + Docker)

All layers chosen for speed-to-demo, compliance surface minimization, and real-time capability.

Architecture — Data Flow



Every layer is purpose-built: audio streams to Deepgram every 3 seconds, transcript chunks feed Claude for structured fact extraction, and the deterministic ranking engine queries Postgres — never an LLM — for sub-100ms carrier decisions.

Core Data Models

All data lives in Supabase Postgres with Row-Level Security (RLS) enforced per agent. Key tables:

contacts

`id, first_name, last_name, phone, email, dob, state, created_by_agent_id`

policies

`id, contact_id, carrier_id, product_type, coverage_amount, premium, status, effective_date`

pipeline_cards / stages

`id, contact_id, stage_id, agent_id, moved_at` + auto_sms_template per stage

carriers

`id, name, products_offered, underwriting_rules_json, quote_formula_json`

agents + commissions

`agents(id, name, email, role, agency_id)` + `commissions(amount, status, expected_date, paid_date)`

call_sessions

`id, agent_id, contact_id, twilio_call_sid, started_at, ended_at`

transcripts

`id, call_session_id, speaker, text, timestamp_ms`

extracted_facts

`id, call_session_id, fact_type, fact_key, fact_value, confidence`

carrier_rankings

`id, call_session_id, carrier_id, rank, score, reason`

quotes

`id, call_session_id, carrier_id, monthly_premium, annual_premium`

Carrier Knowledge Base Schema

Each carrier is a structured JSON file powering the deterministic ranking engine. MVP includes **5 carriers**: Mutual of Omaha, Banner Life, Protective Life, Pacific Life, and National Life Group.

Products Offered

Term, Whole Life, IUL, Final Expense — per carrier

Underwriting Rules

Diabetes (type, max A1C, meds), tobacco quit years, BMI limits, age limits, max coverage

Rate Tables

Per product, per rate class, sample premiums — used for deterministic quoting

Carrier JSON files are the source of truth for all ranking decisions. No LLM touches underwriting logic.

```
{
  "carrier_id": "mutual_of_omaha",
  "products": ["term","whole","final_expense"],
  "underwriting_rules": {
    "diabetes": {
      "type2_max_a1c": 8.5,
      "insulin_allowed": false
    },
    "tobacco": {
      "non_smoker_quit_years": 3
    },
    "bmi_max": 40,
    "age_max": 80
  }
}
```

AI Extraction Tool Schema

CLAUDE TOOL-USE | STREAMING

Tool name: `extract_underwriting_facts` — runs on every 3-second transcript chunk during a live call.

1

Live Transcript Input

Raw spoken text from Deepgram streaming, chunked every 3 seconds

2

Claude Extraction

Extracts: age, gender, state, coverage amount/type, conditions, tobacco status, medications, family history, height, weight

3

Confidence Tagging

Each fact tagged: **confirmed** / **indicated** / **possible**

4

Quote Engine Gate

Only **CONFIRMED** facts reach the ranking engine — prevents hallucination liability

⊗ Indicated and possible facts are displayed to the agent for follow-up but never used in automated underwriting decisions.

MVP Demo Scope — The 90-Second Demo

Every engineering decision is optimized to make this demo land perfectly. Eight beats, ninety seconds.

01

Agent Opens Cockpit

Sees lead in pipeline, ready to dial

02

Clicks "Dial"

Browser softphone rings out via Twilio Voice SDK

03

Client Mentions Age + Coverage

Carrier shortlist populates automatically in real-time

04

"I have Type 2 Diabetes"

Coaching panel injects targeted follow-up questions instantly

05

Client Answers

Fact panel updates, carriers RE-RANK live with animations

06

Smoking History Mentioned

Another re-rank fires — carriers shift in real-time

07

Call Ends

Quote comparison renders: top 3 carriers + premiums + reasons

08

Pipeline Card Moved

SMS fires automatically to client — workflow complete

3-Week Build Schedule

1

Week 1 — Voice Pipe

Day 1: Next.js + Supabase scaffold, auth, dark theme, 3-stage pipeline
Day 2: Twilio Voice SDK browser softphone (outbound dial)
Day 3: Deepgram streaming STT wired to call, live captions
Day 4: Claude streaming tool-use, fact extraction schema
Day 5: Carrier JSON knowledge base, ranking engine unit tests

2

Week 2 — Moat Layer

Day 6: Coaching panel with condition-triggered follow-ups
Day 7: Live carrier re-ranking with animations + reason tags
Day 8: Post-call quote comparison screen
Day 9: Pipeline stage-change SMS trigger
Day 10: Dashboard shell (mock KPIs, commissions, leaderboard)

3

Week 3 — Polish + Demo

Day 11–12: Visual polish, motion, typography
Day 13: Seed demo data, end-to-end dry run
Day 14: Record backup demo video + fallback
Day 15: Buffer day for fixes

What We're NOT Building (MVP)

Scope discipline is critical. The following are explicitly deferred to keep the MVP focused and the demo clean.

→ Real Carrier API Integrations

Using static JSON rules for MVP — live API connections deferred

→ Parallel / Predictive Dialer

Single-line only for MVP — parallel dialing deferred due to TCPA risk

→ Commission Reconciliation

No carrier SFTP ingestion — mock data only for demo

→ E-Signature / DocuSign

Integration deferred to post-MVP phase

→ ACORD / COI Generation

Certificate of Insurance generation requires ACORD licensing — deferred

→ SOC2 Type II

Compliance advisor starts in parallel — certification deferred

→ Full Campaign Management

Lead list → dial campaign automation deferred post-MVP

→ Mobile App + Calendar Sync

Both deferred — desktop-first for MVP demo

Compliance Requirements

TCPA

Provable per-number consent, DNC scrubbing, recording disclosures per state. **2-party states:** CA, FL, IL, MD, MA, MT, NH, PA, WA

TSR

Federal Telemarketing Sales Rules compliance required across all outbound dialing activity

HIPAA-Adjacent

Capturing A1C, diagnoses, prescriptions = treat as PHI. Column-level encryption, audit logs, retention policies required

E&O Defensibility

All ranking logic deterministic and auditable. No RAG, no hallucination in underwriting decisions. Agent selects from AI-ranked options.

⚠ Carrier contracts have redistribution restrictions — legal signoff required before any carrier data is shared externally.

RAG / Knowledge Architecture

CRITICAL DESIGN DECISION

The Rule

DO NOT use RAG for real-time underwriting decisions.

Real-time ranking must be deterministic, sub-100ms, and legally auditable. RAG introduces hallucination risk that is unacceptable in an E&O context.

⊗ No Neo4j needed. Postgres with foreign keys IS the knowledge graph.

MVP (Now)

Deterministic TypeScript rules engine. Queries structured Postgres tables. Sub-100ms. Zero hallucination risk.

Phase 2

pgvector for sales scripts, objection handling, historical transcript search — RAG is safe here, no underwriting decisions.

Phase 3

RAG over 300-page carrier PDFs → human review → promoted to structured rules tables. Human in the loop always.

Key Engineering Decisions

Seven foundational decisions that define the architecture and competitive position of NexusOS.

1

Deterministic Rules for Underwriting

NOT RAG — every carrier ranking decision is auditable TypeScript logic

2

Supabase Over Custom Backend

RLS, Realtime, pgvector, and Edge Functions out of the box — massive velocity advantage

3

Browser Softphone

Twilio Voice SDK in-browser — no hardware phones, no desktop app, zero install friction

4

Claude Tool-Use for Extraction

Not fine-tuned models — tool-use is faster to iterate, auditable, and requires no training data

5

Dark Theme, Insurance-Pro Aesthetic

Signals seriousness and differentiates from consumer-grade CRM tools

6

Internal Tool First

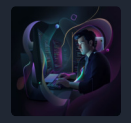
Reduces compliance surface area — product-ize later after legal framework is established

7

Single-Line Dialer First

Parallel dialing deferred — TCPA risk is too high for MVP without dedicated compliance infrastructure

Team & Roles



Zazu — Technical Co-founder / AI + Product

Dialer architecture, transcription pipeline, Claude orchestration, and overall product vision. The driving force behind the AI moat.

TBD — Senior Backend Engineer

Carrier integrations, SOC2 architecture design. Candidate: Zach's brother. Critical hire for compliance infrastructure.

TBD — Senior Frontend Engineer

GHL-level polish, real-time dashboards, dialer UI. Must be comfortable with Framer Motion and complex real-time state.

Fractional — Compliance Advisor

TCPA, state insurance regulations, PHI handling, DNC compliance. Starts in parallel with MVP build.

Getting Started — Environment Setup

Each engineer creates their **OWN** API keys. Never share keys across developers.

Required Environment Variables

Supabase

`NEXT_PUBLIC_SUPABASE_URL`
`ANON_KEY` + `SERVICE_ROLE_KEY`
→ supabase.com/dashboard

Twilio

`TWILIO_ACCOUNT_SID` + `AUTH_TOKEN`
`PHONE_NUMBER` + `TWIML_APP_SID`
→ console.twilio.com

Deepgram

`DEEPGRAM_API_KEY`
→ console.deepgram.com

Anthropic

`ANTHROPIC_API_KEY`
→ console.anthropic.com

Setup Commands

```
git clone https://github.com/yenscastro/insurance
npm install
cp .env.example .env.local
# Fill in your keys
npx supabase start
npm run dev
```

✓ Five commands from clone to running local dev environment.

Repository

github.com/yenscastro/insurance

File Structure

```
nexus-os/
├── app/                # Next.js App Router
│   ├── (auth)/         # Login / signup
│   ├── (dashboard)/
│   │   ├── pipeline/
│   │   ├── cockpit/
│   │   ├── contacts/
│   │   ├── dashboard/
│   │   ├── campaigns/
│   │   ├── quotes/
│   │   └── settings/
│   └── api/
│       ├── twilio/
│       ├── deepgram/
│       ├── ai/
│       └── carriers/
├── components/
│   ├── pipeline/
│   ├── cockpit/
│   ├── dashboard/
│   └── ui/
├── lib/
│   ├── carriers/
│   ├── twilio/
│   ├── deepgram/
│   ├── ai/
│   └── supabase/
├── supabase/migrations/
├── data/carriers/      # JSON carrier knowledge base
└── package.json
```

Architecture Principles

- **App Router First**
  All routes use Next.js 15 App Router with React Server Components where possible
- **Carrier Data as Code**
  JSON files in `data/carriers/` are the single source of truth — versioned in git
- **Lib Layer Isolation**
  All third-party SDK calls isolated in `lib/` — swap vendors without touching UI
- **Migrations Tracked**
  All schema changes via `supabase/migrations/` — no manual SQL in production

Repo: github.com/yenscastro/insurance